

Qualification Pack



Conventional Turning Operator

QP Code: CSC/Q0110

Version: 4.0

NSQF Level: 3

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CSC/Q0110: Conventional Turning Operator

Brief Job Description

A Conventional Turning Operator is responsible for carrying out turning operations using a conventional turning machine. The individual carries out turning operations as per the given instructions to ensure the machined components have the required features and specifications. The person also carries out minor repair and maintenance of the turning machine.

Personal Attributes

The individual must be physically fit to work for long durations with concentration. The person must have basic communication, numerical and problem-solving skills, with the ability to work in coordination with others to achieve the work objectives

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CSC/N1335: Follow the health and safety practices at the work](#)
2. [CSC/N1336: Coordinate with co-workers to achieve work efficiency](#)
3. [CSC/N0110: Operate conventional turning machine to machine a variety of components](#)
4. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	Machine Tools
Occupation	Machining
Country	India
NSQF Level	3
Credits	11
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7223.1201

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Minimum Educational Qualification & Experience	10th Class (Pass) OR Grade 8 pass with 2 years of (NTC/ NAC) after 8th (or equivalent) OR 9th Class (Pass) with 1.5 years of experience relevant OR 8th Class pass with 3 Years of experience relevant OR Previous relevant Qualification of NSQF Level 2.5 with 1.5 years of experience relevant OR Previous relevant Qualification of NSQF Level 2 with 3 Years of experience relevant
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	01/10/2028
NSQC Approval Date	01/10/2025
Version	4.0
Reference code on NQR	QG-03-CG-00179-2023-V1.1-CGSC
NQR Version	4

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CSC/N1335: Follow the health and safety practices at the work

Description

This OS unit is about following the appropriate health and safety practices at work. It covers responsibilities towards self and others to ensure a safe work environment.

Scope

The scope covers the following :

- Maintain personal health and safety
- Assist in hazard management
- Check the first aid box, firefighting and safety equipment
- Assist in waste management
- Follow the fire safety guidelines
- Follow the emergency and first-aid procedures
- Carry out relevant documentation and review

Elements and Performance Criteria

Maintain personal health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** follow the recommended practices to ensure protection from infections and transmission to others, such as the use of hand sanitiser and face mask
- PC2.** check the work conditions, assess the potential health and safety risks, and take appropriate measures to mitigate them
- PC3.** select and use the appropriate Personal Protective Equipment (PPE) relevant to the task and work conditions
- PC4.** follow the recommended techniques while lifting and moving heavy objects to avoid injury
- PC5.** follow the manufacturer's instructions and workplace safety guidelines while working on heavy machinery, tools and equipment

Assist in hazard management

To be competent, the user/individual on the job must be able to:

- PC6.** identify existing and potential hazards at work
- PC7.** assess the potential risks and injuries associated with the identified hazards
- PC8.** coordinate with the supervisor or other relevant personnel to prevent or minimise the identified hazards
- PC9.** handle hazardous materials safely and store them in the designated storage

Check the first aid box, firefighting and safety equipment

To be competent, the user/individual on the job must be able to:

- PC10.** check the first aid box to ensure it is updated with the relevant first aid supplies
- PC11.** check and test the firefighting and various safety equipment to ensure they are in usable condition

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PC12. coordinate with the supervisor for the repair and replacement of firefighting and safety equipment

Assist in waste management

To be competent, the user/individual on the job must be able to:

PC13. segregate waste into appropriate categories

PC14. recycle the recyclable waste appropriately

PC15. dispose of the non-recyclable waste in an environment-friendly manner, complying with the applicable regulations

Follow the fire safety guidelines

To be competent, the user/individual on the job must be able to:

PC16. use the appropriate type of fire extinguisher to extinguish different types of fires safely

PC17. follow the recommended practices for a safe rescue during a fire emergency

PC18. coordinate with the fire department to request assistance to extinguish a serious fire

Follow the emergency and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC19. follow the organisational health and safety guidelines during workplace emergencies to ensure own and co-workers' safety

PC20. follow the recommended practices to minimise loss to organisational property during an emergency

PC21. follow the recommended procedure to free a person from electrocution

PC22. administer appropriate first aid to the injured personnel

PC23. perform Cardiopulmonary Resuscitation (CPR) on a potential victim of cardiac arrest

PC24. coordinate with the emergency services to request medical assistance for seriously injured/ill personnel requiring professional medical attention or hospitalisation

Carry out relevant documentation and review

To be competent, the user/individual on the job must be able to:

PC25. carry out appropriate documentation following a health and safety incident at work, including all the required information

PC26. coordinate with the relevant personnel to review health and safety conditions at work regularly or following an incident

PC27. assist in implementing appropriate changes to improve the health and safety conditions at work

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the recommended practices to be followed to ensure protection from infections and transmission to others, such as the use of hand sanitiser and face mask

KU2. the importance and process of checking the work conditions, assessing the potential health and safety risks, and take appropriate measures to mitigate them

KU3. the importance and process of selecting and using the appropriate PPE relevant to the task and work conditions

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- KU4.** the recommended techniques to be followed while lifting and moving heavy objects to avoid injury
- KU5.** the importance of following the manufacturer's instructions and workplace safety guidelines while working on heavy machinery, tools and equipment
- KU6.** the importance and process of identifying existing and potential hazards at work
- KU7.** the process of assessing the potential risks and injuries associated with the various hazards
- KU8.** how to prevent or minimise different types of hazards
- KU9.** how to handle and store hazardous materials safely
- KU10.** the importance of ensuring the first aid box is updated with the relevant first aid supplies
- KU11.** the process of checking and testing the firefighting and various safety equipment to ensure they are in a usable condition
- KU12.** the criteria for segregating waste into appropriate categories
- KU13.** the appropriate methods for recycling the recyclable waste
- KU14.** the process of disposing of the non-recyclable waste safely and the applicable regulations
- KU15.** Use of different types of fire extinguishers to extinguish different types of fires
- KU16.** the recommended practices to be followed for a safe rescue during a fire emergency
- KU17.** how to request assistance from the fire department to extinguish a serious fire
- KU18.** the appropriate practices to be followed during workplace emergencies to ensure safety and minimise loss to organisational property
- KU19.** common health and safety hazards present in a work environment, associated risks, and how to mitigate them
- KU20.** safe working practices to be followed while working at various hazardous sites and using electrical equipment
- KU21.** the importance of ensuring easy access to firefighting and safety equipment
- KU22.** the appropriate preventative and remedial actions to be taken in the case of exposure to toxic materials, such as poisonous chemicals and gases
- KU23.** various causes of fire in different work environments and the recommended precautions to be taken to prevent fire accidents
- KU24.** different methods of extinguishing fire
- KU25.** different materials used for extinguishing fire, such as sand, water, foam, CO2, dry powder, etc.
- KU26.** the applicable rescue techniques to be followed during a fire emergency
- KU27.** the importance of placing safety signs and instructions at strategic locations in a workplace and following them
- KU28.** different types of first aid treatment to be provided for different types of injuries
- KU29.** potential injuries associated with incorrect manual handling
- KU30.** how to move an injured person safely
- KU31.** various hazards associated with the use of various machinery, tools, implements, equipment and materials
- KU32.** the importance of ensuring no obstruction and free access to fire exits
- KU33.** how to free a person from electrocution safely
- KU34.** how to administer appropriate first aid to an injured person

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- KU35.** how to perform Cardiopulmonary Resuscitation (CPR)
- KU36.** the importance of coordinating with the emergency services to request urgent medical assistance for persons requiring professional medical attention or hospitalisation
- KU37.** the appropriate documentation to be carried out following a health and safety incident at work, and the relevant information to be included
- KU38.** the importance and process of reviewing the health and safety conditions at work regularly or following an incident
- KU39.** the importance and process of implementing appropriate changes to improve the health and safety conditions at work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** communicate clearly and politely with co-workers and clients
- GS3.** read the relevant literature to get the latest updates about the field of work
- GS4.** listen attentively to understand the information being shared
- GS5.** plan and prioritise tasks to ensure timely completion
- GS6.** take quick decisions to deal with workplace emergencies and accidents
- GS7.** identify possible disruptions to work and take appropriate preventive measures
- GS8.** coordinate with the co-workers to achieve the work objectives
- GS9.** evaluate all possible solutions to a problem to select the best one

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal health and safety</i>	7	12	-	-
PC1. follow the recommended practices to ensure protection from infections and transmission to others, such as the use of hand sanitiser and face mask	2	3	-	-
PC2. check the work conditions, assess the potential health and safety risks, and take appropriate measures to mitigate them	1	2	-	-
PC3. select and use the appropriate Personal Protective Equipment (PPE) relevant to the task and work conditions	1	2	-	-
PC4. follow the recommended techniques while lifting and moving heavy objects to avoid injury	1	3	-	-
PC5. follow the manufacturer's instructions and workplace safety guidelines while working on heavy machinery, tools and equipment	2	2	-	-
<i>Assist in hazard management</i>	4	10	-	-
PC6. identify existing and potential hazards at work	1	1	-	-
PC7. assess the potential risks and injuries associated with the identified hazards	1	3	-	-
PC8. coordinate with the supervisor or other relevant personnel to prevent or minimise the identified hazards	1	3	-	-
PC9. handle hazardous materials safely and store them in the designated storage	1	3	-	-
<i>Check the first aid box, firefighting and safety equipment</i>	3	7	-	-
PC10. check the first aid box to ensure it is updated with the relevant first aid supplies	1	2	-	-
PC11. check and test the firefighting and various safety equipment to ensure they are in usable condition	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. coordinate with the supervisor for the repair and replacement of firefighting and safety equipment	1	2	-	-
<i>Assist in waste management</i>	3	8	-	-
PC13. segregate waste into appropriate categories	1	3	-	-
PC14. recycle the recyclable waste appropriately	1	3	-	-
PC15. dispose of the non-recyclable waste in an environment-friendly manner, complying with the applicable regulations	1	2	-	-
<i>Follow the fire safety guidelines</i>	3	12	-	-
PC16. use the appropriate type of fire extinguisher to extinguish different types of fires safely	1	4	-	-
PC17. follow the recommended practices for a safe rescue during a fire emergency	1	4	-	-
PC18. coordinate with the fire department to request assistance to extinguish a serious fire	1	4	-	-
<i>Follow the emergency and first-aid procedures</i>	7	12	-	-
PC19. follow the organisational health and safety guidelines during workplace emergencies to ensure own and co-workers' safety	1	2	-	-
PC20. follow the recommended practices to minimise loss to organisational property during an emergency	1	3	-	-
PC21. follow the recommended procedure to free a person from electrocution	1	2	-	-
PC22. administer appropriate first aid to the injured personnel	1	2	-	-
PC23. perform Cardiopulmonary Resuscitation (CPR) on a potential victim of cardiac arrest	1	2	-	-
PC24. coordinate with the emergency services to request medical assistance for seriously injured/ ill personnel requiring professional medical attention or hospitalisation	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out relevant documentation and review</i>	3	9	-	-
PC25. carry out appropriate documentation following a health and safety incident at work, including all the required information	1	3	-	-
PC26. coordinate with the relevant personnel to review health and safety conditions at work regularly or following an incident	1	3	-	-
PC27. assist in implementing appropriate changes to improve the health and safety conditions at work	1	3	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N1335
NOS Name	Follow the health and safety practices at the work
Sector	Capital Goods
Sub-Sector	Machine Tools, Process Plant Machinery, Dies, Moulds and Press Tools, Electrical and Power Machinery, Plastics Manufacturing Machinery, Light Engineering Goods, Textile Manufacturing Machinery
Occupation	Machining
NSQF Level	3
Credits	TBD
Version	2.0
Last Reviewed Date	01/10/2025
Next Review Date	01/10/2028
NSQC Clearance Date	01/10/2025

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CSC/N1336: Coordinate with co-workers to achieve work efficiency

Description

This OS unit is about working in coordination with co-workers to achieve the work objectives efficiently. It also covers practising inclusion at work.

Scope

The scope covers the following :

- Work effectively with co-workers
- Communicate effectively with co-workers
- Practice inclusion at work

Elements and Performance Criteria

Work effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC1.** plan daily tasks at work to ensure their timely completion and efficient use of time
- PC2.** carry out work responsibilities adhering to the limits of authority
- PC3.** follow the supervisor's instructions to ensure adherence to the applicable quality standards and timescales
- PC4.** coordinate with the co-workers to achieve the work objectives efficiently
- PC5.** prepare the relevant documents and reports as per the supervisor's instructions, providing appropriate information clearly and systematically
- PC6.** coordinate with the supervisor or relevant personnel to deal with out of authority tasks and concerns
- PC7.** mentor and assist subordinates in the execution of their work responsibilities
- PC8.** identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures
- PC9.** use various resources efficiently to ensure maximum utilisation and minimum wastage
- PC10.** follow the recommended practices to avoid and resolve conflicts at work
- PC11.** follow the relevant organisational policies to ensure disciplined behaviour with maximum productivity at work

Communicate effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC12.** follow the organisational policy for the efficient and timely dissemination of information to the authorised personnel
- PC13.** communicate clearly and politely to ensure effective communication with co-workers
- PC14.** follow the appropriate techniques for active listening during interactions

Practice inclusion at work

To be competent, the user/individual on the job must be able to:

- PC15.** empathise with Persons with Disabilities (PWD)

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PC16. adopt gender-neutral behaviour at work

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance and process of effective communication in the workplace
- KU2.** the barriers to effective communication and how to overcome them
- KU3.** the importance of teamwork in an organisation's and individual's success
- KU4.** the importance of active listening in the work environment
- KU5.** the appropriate techniques to be followed for active listening
- KU6.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU7.** the importance of maintaining discipline and ethical behaviour at work
- KU8.** the common reasons for interpersonal conflict and how to resolve them
- KU9.** the importance of developing effective work relationships for professional success
- KU10.** how expressing and addressing grievances appropriately and effectively
- KU11.** the importance and process of planning daily tasks to ensure their timely completion and efficient use of time
- KU12.** the importance of adhering to the limits of authority at work
- KU13.** the importance of following the applicable quality standards and timescales at work
- KU14.** the importance of coordinating with the co-workers to achieve the work objectives efficiently
- KU15.** the relevant documentation requirements
- KU16.** the importance of providing appropriate information clearly and systematically in work documents
- KU17.** the escalation matrix to be followed to deal with out of authority tasks and concerns
- KU18.** the importance and process of mentoring and assisting subordinates in the execution of their work responsibilities
- KU19.** how to identify possible disruptions to work prevent them
- KU20.** how to use various resources efficiently to ensure maximum utilisation and minimum wastage
- KU21.** the recommended practices to be followed at work to avoid and resolve conflicts at work
- KU22.** the importance and process of efficient and timely dissemination of information to the authorised personnel
- KU23.** how to communicate clearly and politely to ensure effective communication
- KU24.** the importance of following the recommended practices to ensure an inclusive environment for PwD and all genders at work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records

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- GS2.** read work-related and other relevant literature
- GS3.** communicate politely and -professionally
- GS4.** listen attentively to understand the information or instructions being shared
- GS5.** plan and prioritise tasks to ensure timely completion
- GS6.** take prompt decisions to deal with workplace emergencies and accidents
- GS7.** evaluate all possible solutions to a problem to select the best one

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with co-workers</i>	20	43	-	-
PC1. plan daily tasks at work to ensure their timely completion and efficient use of time	2	4	-	-
PC2. carry out work responsibilities adhering to the limits of authority	2	4	-	-
PC3. follow the supervisor's instructions to ensure adherence to the applicable quality standards and timescales	2	4	-	-
PC4. coordinate with the co-workers to achieve the work objectives efficiently	2	4	-	-
PC5. prepare the relevant documents and reports as per the supervisor's instructions, providing appropriate information clearly and systematically	2	4	-	-
PC6. coordinate with the supervisor or relevant personnel to deal with out of authority tasks and concerns	2	4	-	-
PC7. mentor and assist subordinates in the execution of their work responsibilities	2	4	-	-
PC8. identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures	2	4	-	-
PC9. use various resources efficiently to ensure maximum utilisation and minimum wastage	2	4	-	-
PC10. follow the recommended practices to avoid and resolve conflicts at work	1	4	-	-
PC11. follow the relevant organisational policies to ensure disciplined behaviour with maximum productivity at work	1	3	-	-
<i>Communicate effectively with co-workers</i>	6	15	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. follow the organisational policy for the efficient and timely dissemination of information to the authorised personnel	2	5	-	-
PC13. communicate clearly and politely to ensure effective communication with co-workers	2	5	-	-
PC14. follow the appropriate techniques for active listening during interactions	2	5	-	-
<i>Practice inclusion at work</i>	4	12	-	-
PC15. empathise with Persons with Disabilities (PwD)	2	6	-	-
PC16. adopt gender-neutral behaviour at work	2	6	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CSC/N1336
NOS Name	Coordinate with co-workers to achieve work efficiency
Sector	Capital Goods
Sub-Sector	Machine Tools, Dies, Moulds and Press Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Light Engineering Goods
Occupation	Machining
NSQF Level	3
Credits	TBD
Version	2.0
Last Reviewed Date	01/10/2025
Next Review Date	01/10/2028
NSQC Clearance Date	01/10/2025

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CSC/N0110: Operate conventional turning machine to machine a variety of components

Description

This OS unit is about setting up a conventional turning machine for use and operating it to carry out turning operations on a variety of components as per the given specifications. It also covers resource optimisation.

Scope

The scope covers the following :

- Prepare for carrying out turning operations
- Carry out turning operations
- Use resources optimally

Elements and Performance Criteria

Prepare for carrying out turning operations

To be competent, the user/individual on the job must be able to:

- PC1.** check the calibration date of all the measuring equipment to ensure it has not expired
- PC2.** determine the job specifications/ machining requirements by referring to the valid and approved documents, such as job card, operational drawings, approved sketches/illustrations, work drawings, assembly drawings, etc.
- PC3.** extract the relevant information such as tapping sizes and threads; feeds and speeds; component ratings; machining symbols and tolerances from reference charts, tables, graphs
- PC4.** check the process sheet and match it with the received drawings and other specifications
- PC5.** select the appropriate tools, equipment and measuring instruments to be used during the turning operations, and check them to ensure they are not worn out or damaged
- PC6.** check the availability of the required raw material for carrying out turning operations
- PC7.** check the raw material/ components to be machined to ensure they are free from foreign objects, dirt and contamination and have the recommended surface texture, limits and tolerances according to the process sheet
- PC8.** check the post-machining sheet to determine if the component meets the applicable quality standards from the previous machining operations
- PC9.** prepare the work area for turning operations, ensuring there are no hazards in the area
- PC10.** plan to carry out the required turning activities and the sequence of operations as per the specifications
- PC11.** follow the recommended safe working practices
- PC12.** check the runout of chuck and the taper of the tailstock
- PC13.** coordinate with the machine setter to check the machine is ready for use
- PC14.** prepare for carrying out turning operations by mounting, positioning and setting a range of work holding devices and cutting tools, such as chuck, faceplate, centre drills, twist/core drills, reamers, etc.

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PC15. position and secure the work holding devices to the machine spindle

PC16. secure the workpiece/ component in a chuck or work holding device using the appropriate fixture, ensuring no distortion

Carry out turning operations

To be competent, the user/individual on the job must be able to:

PC17. set the machine tool speeds and feeds to achieve the given component specifications

PC18. operate the machine tool controls safely as per the manufacturer's instructions, both in hand and power modes

PC19. identify abnormal noises coming from the machine and component, and adjust the feed and Revolutions Per Minute (RPM), as required

PC20. check the condition of the tools being used and repair them, as required

PC21. follow the recommended process to stop and restart the turning machine during emergencies

PC22. check the geometrical accuracies of components, such as diameter on multiple points, taper, surface finishing, groove depth and width, drill depth, Outer Diameter (OD)/Inner Diameter (ID), threading quality and dimensions

PC23. check for the permitted grinding margin on OD/ID

PC24. use lathes and other relevant accessories such as the saddle, capstan/turret head, compound slide, tailstock, taper turning attachments, profile attachments, fixed and travelling steadies, as per the requirement

PC25. carry out turning operations using the relevant equipment, such as solid high-speed tooling, brazed tip tooling, interchangeable tipped tooling to produce components with various features, such as surface texture and finish, breadth, width, length, etc.

PC26. follow the recommended practices to overcome the effects of backlash in machine slides and screws

PC27. carry out roughing and finishing appropriately to achieve the required surface finish and dimensional accuracy

PC28. follow the recommended practices to minimise tool wear

PC29. use the appropriate cutting fluids for different types of materials, such as steel, stainless steel, aluminium, aluminium alloys, copper/copper alloys, cast iron, etc.

PC30. check the diameter, parallelism, surface finish, length, width, concentricity and other critical parameters of the machined components using a range of relevant equipment, such as micrometre, Vernier/digital/dial calliper, dial test indicators (DTI), etc.

PC31. ensure the machined components are free from false tool cuts, burrs and sharp edges

PC32. check the dimensional accuracy by taking a trial cut

PC33. perform the relevant quality checks using the appropriate equipment on the machined components before removing them from the machine

PC34. record the measured values as per the organisational standards and complete the post-machining inspection sheet

PC35. identify inconsistencies in dimensions due to tool wear and correct the offsets accordingly

PC36. adjust various machine settings to maintain the desired accuracy as per the operational requirements

PC37. identify the worn-out and damaged tools and equipment

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- PC38.** sharpen or replace the worn-out/ damaged tools, using the necessary equipment, and modify tool offsets according to the new tools
- PC39.** use the relevant Industry 4.0 manufacturing technologies to ensure interconnectivity, automation, machine learning, and real-time data collection and analysis
- PC40.** follow the relevant safety guidelines while sharpening/ replacing the worn-out/damaged tools
- PC41.** coordinate with the supervisor or other relevant personnel to resolve any complex issues encountered with the turning operations
- PC42.** isolate the turning machine safely after use
- PC43.** carry out regular maintenance of the machine after use, such as cleaning and oiling
- PC44.** collect and dispose of the industrial waste appropriately as per the applicable environmental regulations and organisational procedures
- PC45.** carry out necessary documentation as per the organisational procedure on the completion of machining operations
- PC46.** ensure compliance with the applicable quality standards

Use resources optimally

To be competent, the user/individual on the job must be able to:

- PC47.** optimise the usage of electricity and other resources in various tasks and processes
- PC48.** connect the electrical tools and equipment safely, and turn them off when not in use

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the symbols and conventions appropriate to the applicable ISO standards
- KU2.** the applicable costs, such as production cost, machine hour rate, raw material cost, tool cost, coolant cost, overheads, cycle time, idle time, cost of machine idling, part rejection cost, etc.
- KU3.** the process of selecting cutting tools, tool materials, chip breaker geometry, cutting parameters, etc.
- KU4.** the effect of tool nose radius, speed and feed rate on surface finish
- KU5.** the impact of depth of cut on chatter and surface finish
- KU6.** use of the relevant Personal Protective Equipment
- KU7.** the concepts and benefits of Industry 4.0 and Industrial Internet of Things (IIoT)
- KU8.** how to interpret the component drawings to determine the specification required to achieve through the machining of components
- KU9.** various hazards associated with carrying out turning operations and how to minimise them
- KU10.** the purpose and process of carrying out turning
- KU11.** the process of checking various safety mechanisms available on a conventional turning machine for the correct functioning and using them
- KU12.** how to tighten cam locks or other securing devices appropriately
- KU13.** the importance of checking the process sheet and matching it with the received drawings and other material

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- KU14.** the importance and process of checking the quality of machined components according to the post-machining sheet to ensure conformance to the applicable quality standards
- KU15.** how to run the part program in single block mode and the importance of checking the tool condition after each operation
- KU16.** the importance of maintaining the recommended coolant levels and positioning the coolant nozzles appropriately
- KU17.** the importance of checking the sequence of the program as per the process sheet
- KU18.** the importance of checking for the presence of the appropriate tools in the relevant pocket of Automatic Tool Changer (ATC)
- KU19.** how to identify abnormal noises coming from the machine and component
- KU20.** how to adjust the feed and Revolutions Per Minute (RPM)
- KU21.** the importance of checking the condition of tools being used in machining at appropriate intervals during the process
- KU22.** how to identify inconsistencies in the dimensions due to tool wear and the process of correcting the offsets accordingly
- KU23.** how to adjust the machine settings to maintain the desired accuracy
- KU24.** the process of sharpening or replacing the worn-out/ damaged tools, using the necessary equipment
- KU25.** the importance of modifying the tool offsets according to the new tools replacing them
- KU26.** the relevant safety guidelines to be following while sharpening/ replacing the worn-out/damaged tools
- KU27.** the importance of keeping the work area clean and hazard-free
- KU28.** how to use the metric system of measurement
- KU29.** various features of lathes and use of relevant accessories such as saddle, compound slide, tailstock, taper turning attachments, profile attachments, fixed and travelling stays, etc.
- KU30.** the classification and properties of various tool materials
- KU31.** the factors that affect the selection of cutting feeds, and speed and the depth of cut
- KU32.** use of different types of tooling, such as solid high-speed tooling, brazed tip tooling, interchangeable tipped tooling, indexable insert tooling
- KU33.** use of different types of cutting tools such as recess/groove, chamfer, centre drill, twist/core drill, reamer, thread tool, etc.
- KU34.** various component features that can be produced on metal and non-metal components through turning operations, such as flat face, parallel, tapered, eccentric diameters drilling, reamed, bored holes chamfers, grooves/undercuts, profile form, internal and external thread, parting off, knurls or special finish, etc.
- KU35.** the effects of backlash in machine slides and screws, and how to resolve this
- KU36.** the importance of following the safety instructions and warning signs on the machine
- KU37.** different types of cutting fluids and their properties
- KU38.** the importance of clamping the workpiece in a chuck/work holding device, and how incorrect clamping can cause distortion in the finished components
- KU39.** the relevant equipment to be used for checking critical quality parameters, such as external and internal micrometre, Vernier/digital/dial calliper, dial test indicators (DTI), surface finish equipment, steel rules, depth Vernier, etc.

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- KU40.** various critical parameters to be checked, such as external and internal diameters, eccentricity, parallelism, bore/hole size/fit, angle/taper, surface finish, length, depth, position, width, concentricity, ovality, straightness, squareness, etc
- KU41.** the documentation requirements relevant to the job role
- KU42.** the benefits and methods of resource optimisation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** undertake work-related numerical computations
- GS3.** use the appropriate units of measurement and measuring techniques
- GS4.** read the relevant literature to learn about the latest developments in the field of work
- GS5.** listen attentively to understand the information or instructions being shared
- GS6.** communicate politely and professionally
- GS7.** plan and prioritise tasks to ensure timely completion
- GS8.** coordinate with co-workers to achieve the work objectives
- GS9.** evaluate all possible solutions to a problem to select the best one
- GS10.** take prompt decisions to deal with workplace emergencies and accidents

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for carrying out turning operations</i>	16	20	-	-
PC1. check the calibration date of all the measuring equipment to ensure it has not expired	1	2	-	-
PC2. determine the job specifications/ machining requirements by referring to the valid and approved documents, such as job card, operational drawings, approved sketches/illustrations, work drawings, assembly drawings, etc.	1	1	-	-
PC3. extract the relevant information such as tapping sizes and threads; feeds and speeds; component ratings; machining symbols and tolerances from reference charts, tables, graphs	1	2	-	-
PC4. check the process sheet and match it with the received drawings and other specifications	1	1	-	-
PC5. select the appropriate tools, equipment and measuring instruments to be used during the turning operations, and check them to ensure they are not worn out or damaged	1	2	-	-
PC6. check the availability of the required raw material for carrying out turning operations	1	1	-	-
PC7. check the raw material/ components to be machined to ensure they are free from foreign objects, dirt and contamination and have the recommended surface texture, limits and tolerances according to the process sheet	1	2	-	-
PC8. check the post-machining sheet to determine if the component meets the applicable quality standards from the previous machining operations	1	1	-	-
PC9. prepare the work area for turning operations, ensuring there are no hazards in the area	1	1	-	-
PC10. plan to carry out the required turning activities and the sequence of operations as per the specifications	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow the recommended safe working practices	1	1	-	-
PC12. check the runout of chuck and the taper of the tailstock	1	1	-	-
PC13. coordinate with the machine setter to check the machine is ready for use	1	1	-	-
PC14. prepare for carrying out turning operations by mounting, positioning and setting a range of work holding devices and cutting tools, such as chuck, faceplate, centre drills, twist/core drills, reamers, etc.	1	1	-	-
PC15. position and secure the work holding devices to the machine spindle	1	1	-	-
PC16. secure the workpiece/ component in a chuck or work holding device using the appropriate fixture, ensuring no distortion	1	1	-	-
<i>Carry out turning operations</i>	30	30	-	-
PC17. set the machine tool speeds and feeds to achieve the given component specifications	1	1	-	-
PC18. operate the machine tool controls safely as per the manufacturer's instructions, both in hand and power modes	1	1	-	-
PC19. identify abnormal noises coming from the machine and component, and adjust the feed and Revolutions Per Minute (RPM), as required	1	1	-	-
PC20. check the condition of the tools being used and repair them, as required	1	1	-	-
PC21. follow the recommended process to stop and restart the turning machine during emergencies	1	1	-	-
PC22. check the geometrical accuracies of components, such as diameter on multiple points, taper, surface finishing, groove depth and width, drill depth, Outer Diameter (OD)/Inner Diameter (ID), threading quality and dimensions	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. check for the permitted grinding margin on OD/ID	1	1	-	-
PC24. use lathes and other relevant accessories such as the saddle, capstan/turret head, compound slide, tailstock, taper turning attachments, profile attachments, fixed and travelling steadies, as per the requirement	1	1	-	-
PC25. carry out turning operations using the relevant equipment, such as solid high-speed tooling, brazed tip tooling, interchangeable tipped tooling to produce components with various features, such as surface texture and finish, breadth, width, length, etc.	1	1	-	-
PC26. follow the recommended practices to overcome the effects of backlash in machine slides and screws	1	1	-	-
PC27. carry out roughing and finishing appropriately to achieve the required surface finish and dimensional accuracy	1	1	-	-
PC28. follow the recommended practices to minimise tool wear	1	1	-	-
PC29. use the appropriate cutting fluids for different types of materials, such as steel, stainless steel, aluminium, aluminium alloys, copper/copper alloys, cast iron, etc.	1	1	-	-
PC30. check the diameter, parallelism, surface finish, length, width, concentricity and other critical parameters of the machined components using a range of relevant equipment, such as micrometre, Vernier/digital/dial calliper, dial test indicators (DTI), etc.	1	1	-	-
PC31. ensure the machined components are free from false tool cuts, burrs and sharp edges	1	1	-	-
PC32. check the dimensional accuracy by taking a trial cut	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. perform the relevant quality checks using the appropriate equipment on the machined components before removing them from the machine	1	1	-	-
PC34. record the measured values as per the organisational standards and complete the post-machining inspection sheet	1	1	-	-
PC35. identify inconsistencies in dimensions due to tool wear and correct the offsets accordingly	1	1	-	-
PC36. adjust various machine settings to maintain the desired accuracy as per the operational requirements	1	1	-	-
PC37. identify the worn-out and damaged tools and equipment	1	1	-	-
PC38. sharpen or replace the worn-out/ damaged tools, using the necessary equipment, and modify tool offsets according to the new tools	1	1	-	-
PC39. use the relevant Industry 4.0 manufacturing technologies to ensure interconnectivity, automation, machine learning, and real-time data collection and analysis	1	1	-	-
PC40. follow the relevant safety guidelines while sharpening/ replacing the worn-out/damaged tools	1	1	-	-
PC41. coordinate with the supervisor or other relevant personnel to resolve any complex issues encountered with the turning operations	1	1	-	-
PC42. isolate the turning machine safely after use	1	1	-	-
PC43. carry out regular maintenance of the machine after use, such as cleaning and oiling	1	1	-	-
PC44. collect and dispose of the industrial waste appropriately as per the applicable environmental regulations and organisational procedures	1	1	-	-
PC45. carry out necessary documentation as per the organisational procedure on the completion of machining operations	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC46. ensure compliance with the applicable quality standards	1	1	-	-
<i>Use resources optimally</i>	2	2	-	-
PC47. optimise the usage of electricity and other resources in various tasks and processes	1	1	-	-
PC48. connect the electrical tools and equipment safely, and turn them off when not in use	1	1	-	-
NOS Total	48	52	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CSC/N0110
NOS Name	Operate conventional turning machine to machine a variety of components
Sector	Capital Goods
Sub-Sector	Machine Tools, Dies, Moulds and Press Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Light Engineering Goods
Occupation	Machining
NSQF Level	3
Credits	7
Version	3.0
Last Reviewed Date	01/10/2025
Next Review Date	01/10/2028
NSQF Clearance Date	01/10/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQF Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for the Qualification Pack will be created by CGSC.
2. Performance Criteria (PC) have been assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
3. The assessment for the theory part will/may be based on knowledge bank of questions approved CGSC.
4. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
5. Assessment Agencies will create Assessor Guides comprising of Theory and Practical Assessment Set and Guidelines for each examination/training centre (as per assessment criteria below). The same will be approved by CGSC for adequacy.
6. To successfully attain Certification on the Qualification Pack, the trainee must score a minimum of 70% in each Core NOS and minimum of 70% in all non-core NOS. In addition, a candidate needs to attain a minimum overall pass percentage of 70% for certification.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CSC/N1335.Follow the health and safety practices at the work	30	70	-	-	100	20
CSC/N1336.Coordinate with co-workers to achieve work efficiency	30	70	-	-	100	20
CSC/N0110.Operate conventional turning machine to machine a variety of components	48	52	-	-	100	50
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	128	222	-	-	350	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
CO2	Carbon Dioxide
CPR	Cardiac Pulmonary Resuscitation
PPE	Personal Protective Equipment
CMM	Coordinate Measuring Machine
ISO	International Organization For Standardization
DTI	Dial Test Indicators

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.